

ASSESSMENT TOOL 2 – Concept Mapping Questions

ITA0609 - Machine Learning

Sameer Ahmed G

192421154

Q1. Design a Machine Learning System for Health Analysis

Problem Statement

Design a machine learning system to predict diseases such as **Heart Disease** or **Diabetes** using patient information like **Age, Blood Pressure, Glucose Level, Cholesterol Level, BMI, Lifestyle Factors (Smoking, Exercise), and Family History**. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

Introduction

Machine Learning (ML) plays an important role in modern healthcare by helping doctors detect diseases at an early stage. Traditional diagnosis relies heavily on medical expertise, whereas ML systems analyze historical patient data to identify hidden patterns and make predictions.

Diseases such as heart disease and diabetes are influenced by several factors including age, blood pressure, glucose level, cholesterol, obesity, and lifestyle habits. By training ML models on historical medical records, hospitals can predict disease risk more accurately and provide timely treatment.

The goal of this system is to classify a patient as either **Disease Present** or **Disease Not Present**.

System Design

The machine learning system consists of the following stages:

1. Patient Data Collection
2. Data Preprocessing
3. Feature Selection
4. Model Selection
5. Model Training

6. Model Evaluation

7. Disease Prediction

The system accepts patient health parameters as input and predicts whether the patient is likely to suffer from heart disease or diabetes.

Input Features

The system collects the following patient information:

- Age
- Gender
- Blood Pressure
- Glucose Level
- Cholesterol Level
- Body Mass Index (BMI)
- Smoking Habit
- Alcohol Consumption
- Physical Activity
- Family Medical History
- Heart Rate
- Insulin Level

These attributes become the input features used by the machine learning model.

Data Preprocessing

Before training the model, the collected data must be cleaned and prepared.

1. Handling Missing Values

Some medical records may contain missing values.

Methods:

- Replace missing values with Mean
- Replace with Median
- Remove incomplete records if necessary
-

Example:

Missing Blood Pressure → Replace with Average Blood Pressure.

2. Removing Duplicate Records

Duplicate patient records may bias the model.

Duplicate entries are removed to maintain data quality.

3. Encoding Categorical Data

Machine learning algorithms cannot process text directly.

Examples:

Smoking

Yes → 1

No → 0

Gender

Male → 0

Female → 1

4. Feature Scaling

Medical attributes have different ranges.

Example:

Age = 45

Glucose = 180

Blood Pressure = 120

Scaling ensures that all features contribute equally.

Common techniques:

- Min-Max Normalization
- Standardization (Z-score)

5. Outlier Detection

Extremely abnormal values may reduce prediction accuracy.

Example:

Blood Pressure = 400

Glucose = 900

Such values are detected and corrected or removed.

Feature Selection

Not every attribute contributes equally to disease prediction.

Feature selection helps identify the most important attributes.

Common techniques include:

- Correlation Analysis
- Information Gain
- Chi-Square Test
- Recursive Feature Elimination (RFE)

Example:

For diabetes prediction, the important features may be:

- Glucose Level
- BMI
- Age
- Insulin Level
- Family History

Removing irrelevant features reduces training time and improves accuracy.

Algorithm Selection

Several machine learning algorithms can be used.

1. Decision Tree

A Decision Tree divides data into different branches based on attribute values.

Advantages:

- Easy to understand
- Handles both numerical and categorical data
- Provides clear decision rules

Suitable for medical diagnosis because doctors can easily interpret the decision path.

2. Random Forest

Random Forest combines multiple Decision Trees.

Advantages:

- Higher accuracy
- Reduces overfitting
- Robust against noisy data

Random Forest is widely used in disease prediction because it performs well on medical datasets.

3. Logistic Regression

Used for binary classification problems.

Output:

Disease = Yes

Disease = No

Advantages:

- Simple
- Fast
- Easy to interpret

Suitable when predicting the presence or absence of disease.

4. Support Vector Machine (SVM)

SVM separates healthy and diseased patients using an optimal decision boundary.

Advantages:

- High accuracy
- Works well with high-dimensional medical data

Best Choice

Among these algorithms,

Random Forest is the most suitable because:

- High prediction accuracy
- Handles large datasets
- Resistant to overfitting
- Works well with mixed medical attributes
- Provides feature importance

Model Training

The dataset is divided into:

Training Data → 80%

Testing Data → 20%

Training Steps:

1. Split the dataset.
2. Train the Random Forest classifier.
3. Optimize model parameters.
4. Validate using cross-validation.
5. Test on unseen patient data.

Model Evaluation

The model is evaluated using:

- Accuracy
- Precision
- Recall

- F1-Score
- Confusion Matrix
- ROC Curve

Higher values indicate better prediction performance.

Applications

The proposed system can be used in:

- Hospitals
- Clinics
- Telemedicine
- Health Monitoring Systems
- Wearable Devices
- Preventive Healthcare

Advantages

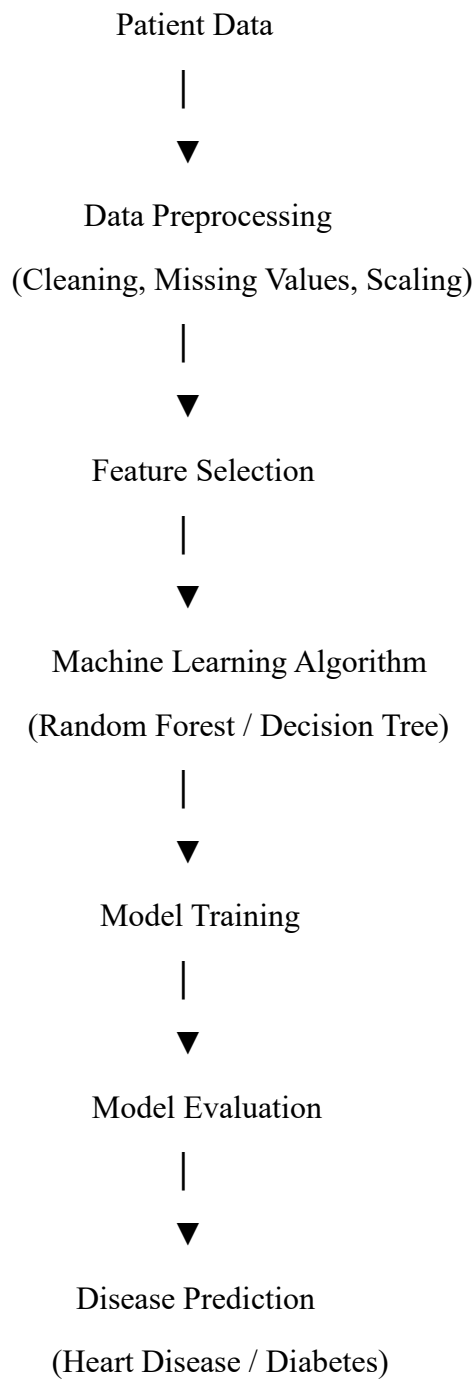
- Early disease detection
- Improved diagnosis
- Faster treatment
- Reduced healthcare costs
- Better patient outcomes

Conclusion

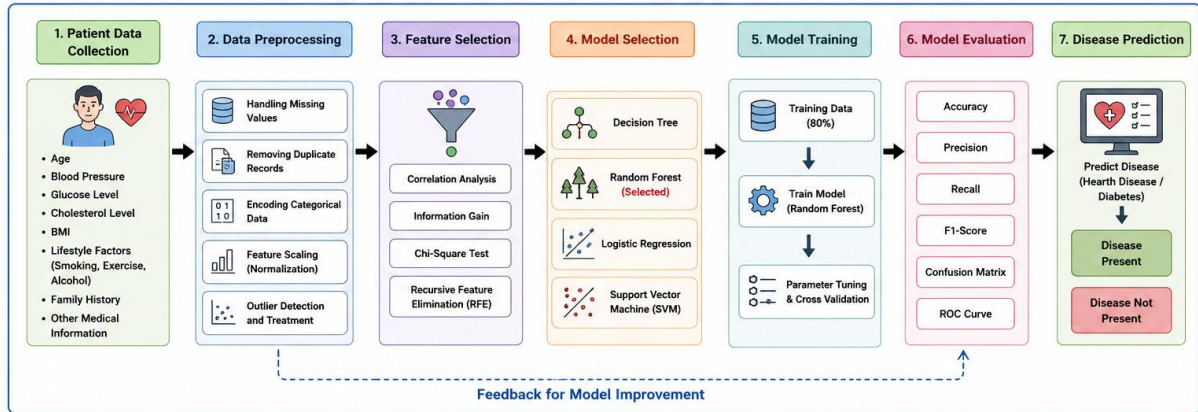
The proposed machine learning system predicts diseases such as heart disease and diabetes using patient health information. After preprocessing and selecting important features, the Random Forest algorithm is trained to classify patients accurately. This system assists healthcare professionals in making reliable and timely medical decisions, ultimately improving patient care.

Diagram(Concept Map / Architecture)

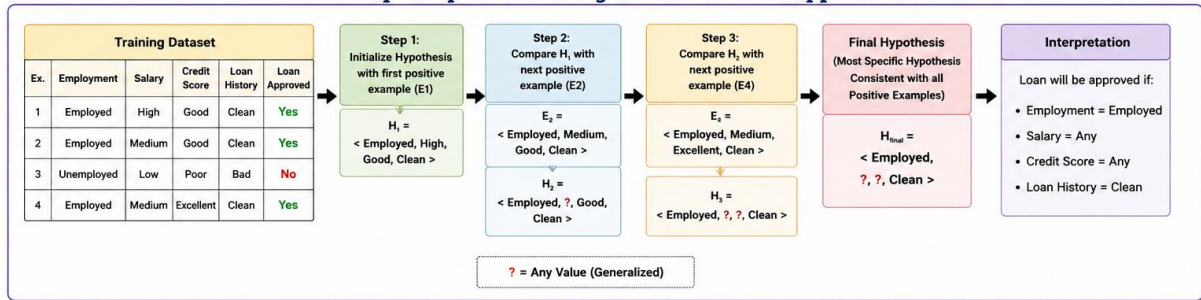
Draw this flowchart in your document:



1. Concept Map – Machine Learning System for Health Analysis



2. Concept Map – Find-S Algorithm for Loan Approval Dataset



Question 2

Apply the Find-S Algorithm

Problem Statement

Apply the Find-S Algorithm on the following dataset to derive the most specific hypothesis consistent with the positive examples.

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Introduction

The Find-S Algorithm is one of the earliest supervised machine learning algorithms. It is used to determine the **most specific hypothesis** that correctly classifies all positive training examples.

The algorithm considers only positive examples ("Yes") and ignores negative examples ("No"). It gradually generalizes the hypothesis until it is consistent with all positive examples.

Positive Examples

Only positive examples are considered:

Example	Employment	Salary	Credit Score	Loan History
1	Employed	High	Good	Clean
2	Employed	Medium	Good	Clean
4	Employed	Medium	Excellent	Clean

Step 1: Initialize Hypothesis

Take the first positive example.

Hypothesis

H₁

<Employed, High, Good, Clean>

Step 2: Compare with Example 2

Example 2

<Employed, Medium, Good, Clean>

Compare each attribute.

Employment

Employed = Employed

Keep Employed.

Salary

High $\neq$ Medium

Replace with ?

Credit Score

Good = Good

Keep Good.

Loan History

Clean = Clean

Keep Clean.

Updated hypothesis

<Employed, ?, Good, Clean>

Step 3: Compare with Example 4

Example 4

<Employed, Medium, Excellent, Clean>

Employment

Same

Keep Employed.

Salary

Already ?

Remain ?

Credit Score

Good $\neq$ Excellent

Replace with ?

Loan History

Clean = Clean

Keep Clean.

Final hypothesis becomes

$\langle \text{Employed}, ?, ?, \text{Clean} \rangle$

Final Hypothesis

The most specific hypothesis generated by Find-S is:

$\langle \text{Employed}, ?, ?, \text{Clean} \rangle$

Interpretation

The hypothesis indicates that:

- Employment must be **Employed**
- Salary can be **any value**
- Credit Score can be **any value**
- Loan History must be **Clean**

Thus, a loan is likely to be approved if the applicant is employed and has a clean loan history, regardless of salary or credit score.

Working of Find-S Algorithm

1. Initialize the hypothesis using the first positive example.
2. Compare the hypothesis with the next positive example.
3. If an attribute differs, replace it with "?".
4. Ignore all negative examples.
5. Repeat until all positive examples are processed.
6. The final hypothesis is the most specific hypothesis consistent with all positive examples.

Advantages

- Simple to implement.
- Easy to understand.
- Efficient for small datasets.
- Produces the most specific consistent hypothesis.

Limitations

- Uses only positive examples.
- Ignores negative examples.
- Sensitive to noisy data.
- Cannot represent multiple hypotheses.

Conclusion

The Find-S algorithm successfully derives the hypothesis **<Employed, ?, ?, Clean>**, indicating that loan approval mainly depends on the applicant being employed and having a clean loan history. Salary level and credit score are generalized because they vary among the positive examples.

Diagram(Find-S Concept Map)

